Taxonomic Classification Analysis of Human Gut Microbiota Using Kraken 2

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ABSTRACT

16S rRNA sequencing has been widely used for identification of the bacterial species. Although, there are various tools for this purpose, they either require long run times or immense computational power that's where the Galaxy platforms comes very handy. Additionally, the Galaxy don't require any prior knowledge of coding, which offers painless usage of the tools.

In our study, we obtained sequences of human gut microbiome within a small community working at a specific location and their families. Stool samples were collected from people who had either recently recovered from Covid-19 infection or had never been infected by Covid-19. All data analysis were done on the Galaxy/Europe platform. Kraken 2 was used for taxonomic classification of bacterial sequences and Python was used to visualize relative abundance within the taxonomic classifications of Kraken 2. Analyses on the Galaxy provided an insight for our study that was quick and easy.

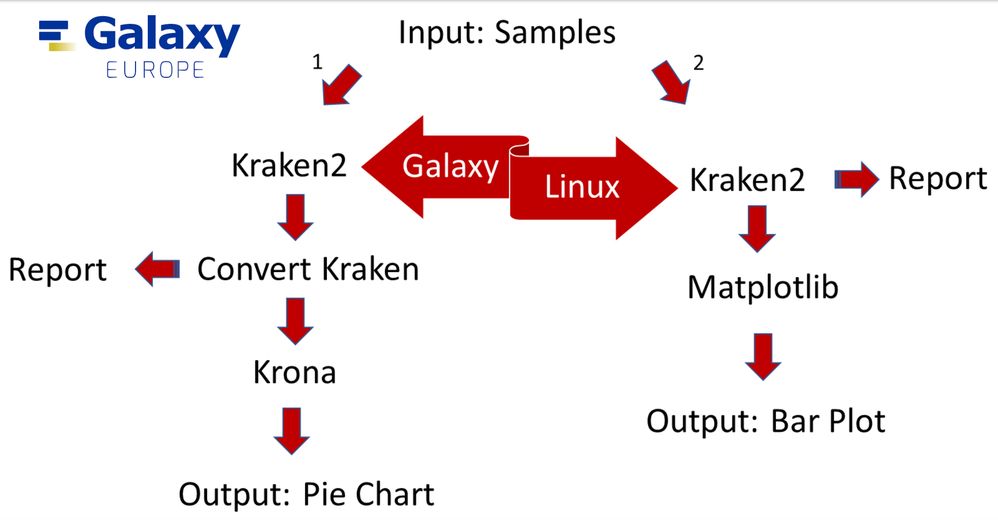
INTRODUCTION

There is growing evidence showing that the association between endosymbionts and the human body varies depending on health condition, lifestyle, diet, antibiotic usage, age, gender and geographical location. Since Covid-19 is a major pandemic facing the world today, investigating its effect on the human gut microbiome is as important as these conditions. Researchers have utilized 16S rRNA sequencing for identification of the bacterial species to understand the structure and function of the microbiome. Improvements in sequencing technology have increased our ability.

Kraken 2 is the most recent version of Kraken, a taxonomic classification system that achieves high accuracy and fast classification speeds by employing exact k-mer matches to metagenomic DNA sequences. It provides different 16 rRNA databases for classification.

The aim of this study was to investigate changes in fecal microbiomes of people who had either recently recovered from COVID-19 infection or had never been infected by COVID-19.

MATERIALS & METHODS



DISCUSSION

We explored whether recovery from Covid-19 infection was associated with restoration of gut microbiome to a community level similar to that of healthy individuals. Overall, frequency of phylum of bacteria no varied significantly between the individuals that recovered from Covid-19 and healthy individuals. On the other hand, *a higher abundance of Actinobacteria* was observed in recovered people from Covid-19 as a result analysis with different databases. It is known that *Actinobacteria* is one of the phyla of gut microbiota in healthy individuals[1]. It suggests that the amount of *Actinobacteria* might increase regarding after virus clearance as an indicator of recovery. The result of classification analysis using Standard Database was shown the significantly increase of *Planctomycetes* in people recovered from Covid-19. *Planctomycetes* have just been associated with human gut as opportunistic pathogens [2-5]. Morever, receiving peptidoglycan inhibitor antibiotics has been shown to result in a high prevalence of *Planctomycetes*[6]. It might be thought that increase of *Planctomycetes* may have occurred with the intake of antibiotics in Covid-19 infected patients.

In our study, we used Minikraken database, Standart database and NCBI Refseq to increase the accuracy of analyses. Comparing the Minikraken and the Standard database, only two phyla have showed significancy (data not shown). Since the Standard database is big and takes longer to run, Minikraken can be considered to provide a faster way for this type of analysis.

Although sample 12 (S12) gave a different classification result compared to the other samples, it did not make a significant difference when the experimental groups were compared. Many factors that may affect the fecal microbiome such as consumption of a product, drug intake, or physical activity, might be considered to influence the sequencing analysis.

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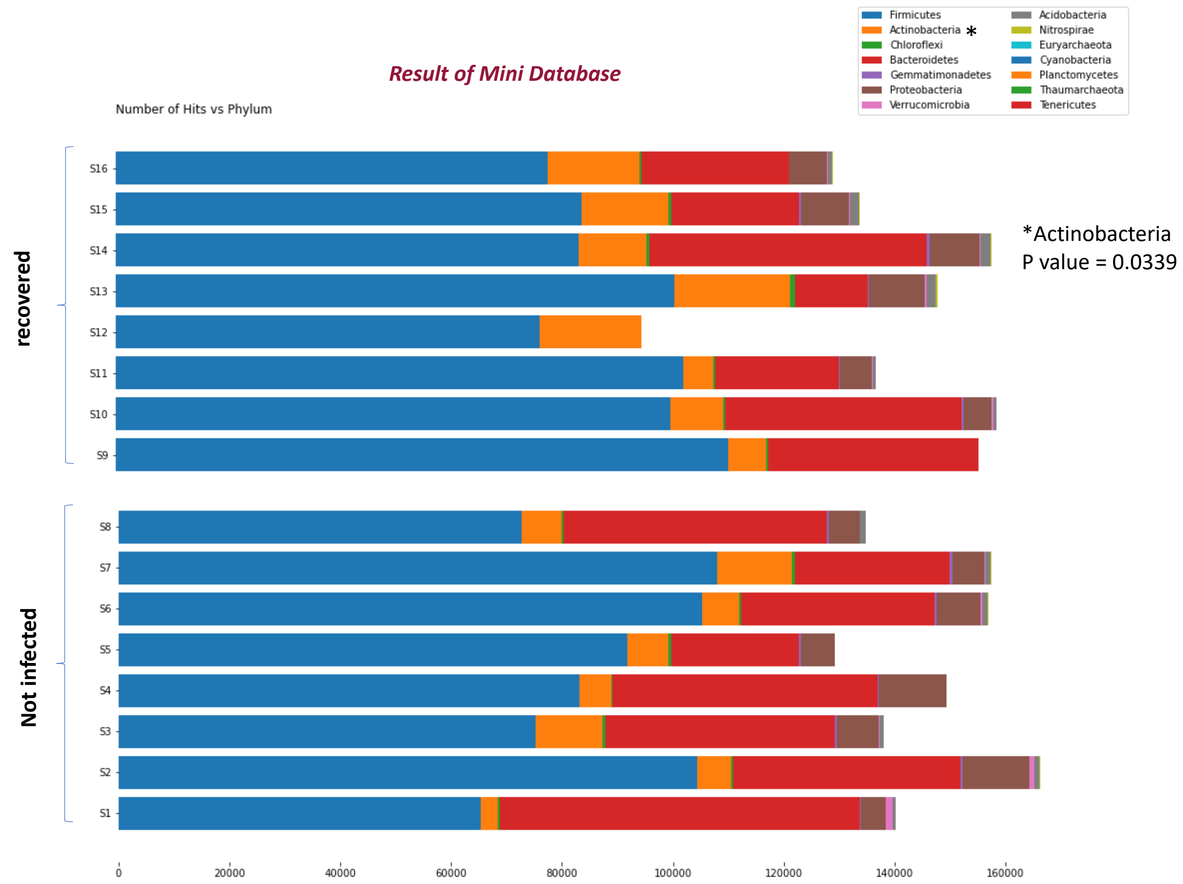
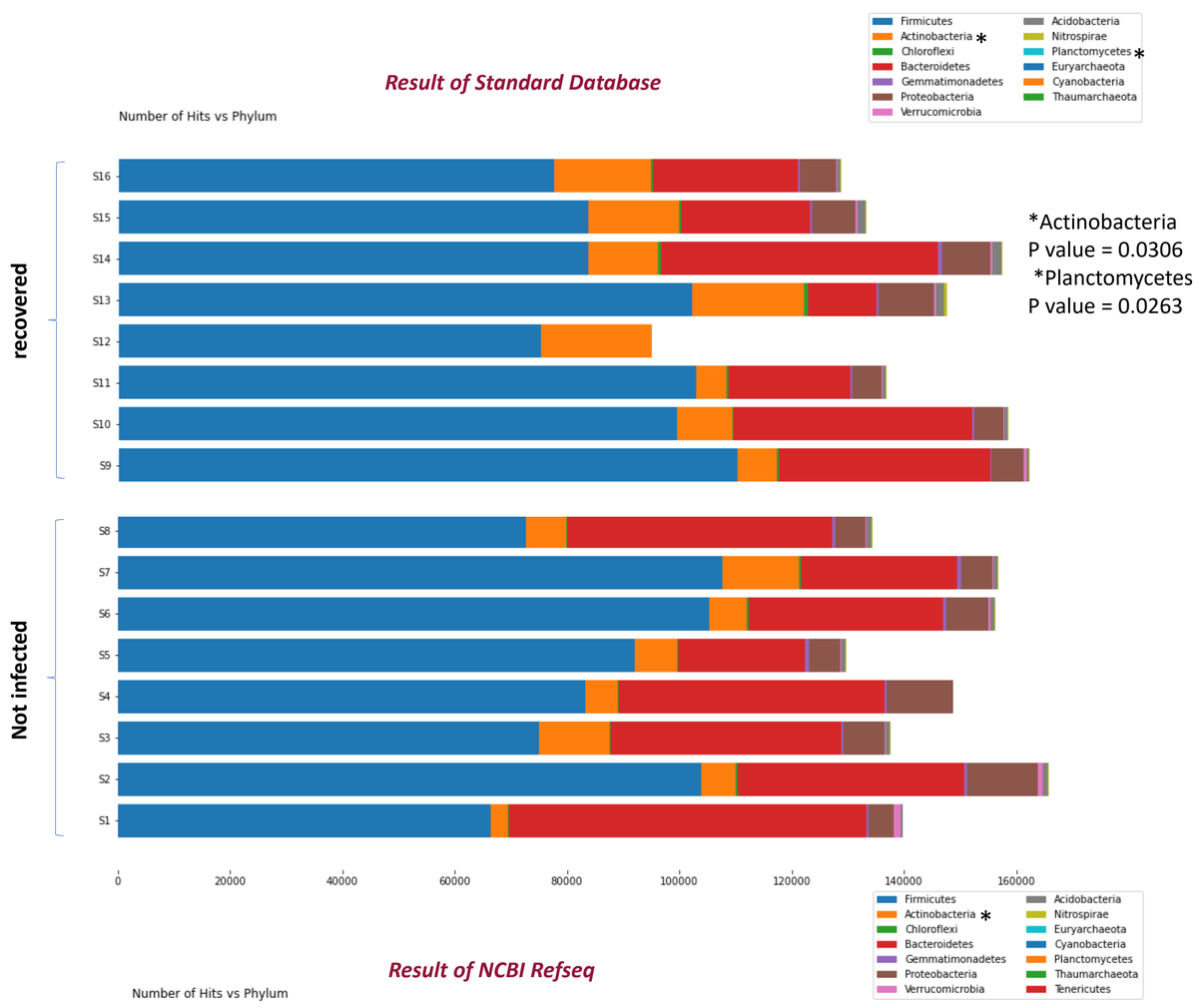
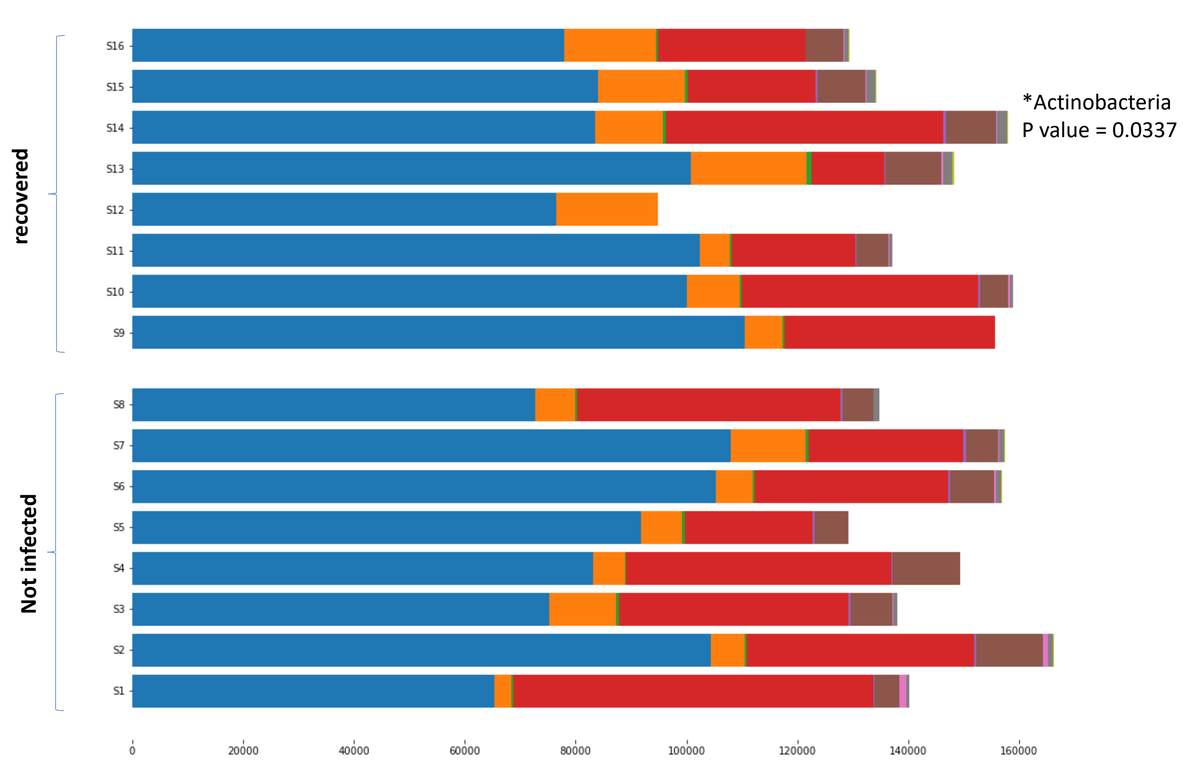
RESULTS

Taxonomic Classification Using Kraken 2







***P ≤ 0.05; Threshold of Number of Hits =100**

- **indicates comporision by phylums in case of had never been infected by COVID-19 and recovered from COVID-19**

S1-S8: not infected people by COVID-19; S9-16: recovered people from COVID-19

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